Zebra puzzle

Five houses in five different colors, inhabited by five people of different nationalities, smoking five different brands of cigarettes and keeping five different pets.

House	House 2	House 3	House 4	House 5
Nationality				
Color				
Pet				
Cigarette				

Clues:

1. The Englishman lives in the red house.
2. The Spaniard owns the dog.
3. Coffee is drunk in the green house.
4. The Ukrainian drinks tea.
5. The green house is immediately to the right of the ivory house.
6. The Old Gold smoker owns snails.
7. Kools are smoked in the yellow house.
8. Milk is drunk in the middle house.
9. The Norwegian lives in the first house.
10. The man who smokes Chesterfields lives in the house next to the man with the fox.
11. Kools are smoked in the house next to the house where the horse is kept.
12. The Lucky Strike smoker drinks orange juice.
13. The Japanese smokes Parliaments.
14. The Norwegian lives next to the blue house.

Rules

1. Each house is its own unique color.
2. All houses are occupied.
3. No two owners have the same pet, smoke the same brand, or drink the same beverage.

DAT Sound Solution

Contents

1. Formulation of the task as a constraint problem
   1.1 Initial state
   1.2 Goal state
   1.3 Operators
   1.4 Constraints
   1.5 Initial assignment of variables
   1.6 Initial representation of variables
   1.7 Initial representation of constraints
   1.8 Initial representation of constraints
   1.9 Initial representation of constraints
   1.10 Initial representation of constraints
   1.11 Initial representation of constraints
   1.12 Initial representation of constraints
   1.13 Initial representation of constraints
   1.14 Initial representation of constraints
   1.15 Initial representation of constraints
   1.16 Initial representation of constraints
   1.17 Initial representation of constraints